

*Note: You may leave arithmetic and trigonometric quantities unsimplified. **No notes. No calculators.** Show all your work and clearly indicate your answer for full credit. Partial solutions will receive partial credit.*

1. (15 points) Consider the differential equation

$$u_{tt} = u_{xx} - u.$$

- (a) Apply separation of variables $u(t, x) = w(t)v(x)$ to write a separated system of differential equations.
- (b) Classify the possible separable solutions $u(t, x) = w(t)v(x)$ in terms of λ .

2. (15 points) The differential equation $u_t = 2u_{xx}$ with $u_x(0, t) = u_x(1, t) = 0$ leads to the system of equations

$$\begin{aligned}w'(t) - 2\lambda w(t) &= 0 \\v''(x) - \lambda v(x) &= 0,\end{aligned}$$

and the boundary conditions $v'(0) = v'(1) = 0$. Find, with justification, the eigenvalues and corresponding eigenfunctions of the system. Find the general solution.

3. (15 points) Consider the function $f(x) = 4x$ on $(-1, 1)$.
- (a) Explain why $f(x) \perp \cos(\pi nx)$ for $n = 1, 2, \dots$
 - (b) Find a Fourier series representation¹ for the function $f(x)$.
 - (c) Can the Fourier series for $f(x)$ be differentiated term-by-term to find a Fourier series for $f'(x)$? Why or why not. Be precise.

¹ $a_n = \frac{1}{\ell} \int_{-\ell}^{\ell} f(x) \cos\left(\frac{\pi nx}{\ell}\right) dx, \quad b_n = \frac{1}{\ell} \int_{-\ell}^{\ell} f(x) \sin\left(\frac{n\pi x}{\ell}\right) dx$

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